

Complete Experimental Results
Offshore Production Monitoring Benchmark

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1 3W Fault Classification

Dataset: Petrobras 3W v2.0.0 — 208,973 sliding windows ($720 \text{ timesteps} \times 27 \text{ sensors}$), compressed to (14×27) statistical feature matrices, 10 classes. Outer split: 80/20 temporal holdout ($n_{\text{test}} = 41,515$). Inner CV: 5-fold stratified.

Table 1: 3W fault classification — held-out test and inner 5-fold CV. Best test values per column in **bold**. FM = foundation model (zero-shot). Tree = Random Forest ensemble. [†]After 30-trial Optuna HPO.

Model	Held-out Test				CV Accuracy
	Acc.	F_1m	F_1w	AUC-PR	mean $\pm$ std
DeepONet	.968	.964	.969	.934	.956 $\pm$.013
LSTM	.967	.962	.967	.931	.951 $\pm$.014
Random Forest ^{Tree}	.966	.964	.966	.985	.979 $\pm$.016
PatchTST [†]	.965	.962	.965	.931	.955 $\pm$.018
FKM-AD (feat.)	.967	.961	.967	.929	.960 $\pm$.019
FKM-AD (raw)	.949	.938	.949	.888	.936 $\pm$.021
ConvTimeNet (feat.) [†]	.966	.962	.966	.930	.955 $\pm$.015
ConvTimeNet (raw)	.158	.027	.043	.100	.631 $\pm$.064
MambaSL (feat.) [†]	.966	.961	.966	.927	.952 $\pm$.012
MambaSL (raw) [†]	.919	.887	.920	.807	.930 $\pm$.010
TiRex ^{FM}	.912	.895	.911	.938	.899 $\pm$.021
<i>Majority baseline</i>	<i>.174</i>	<i>.101</i>	—	—	—

[†]30-trial Optuna HPO. All others use manually tuned hyperparameters or defaults.

Table 2: Optuna HPO results on 3W (30 trials per model; 33 for DeepONet; 5-fold inner CV). *Best CV F_1m* is the objective optimized by Optuna. *Retrain* metrics are from a model retrained on the full training pool with best hyperparameters and evaluated on the held-out test set.

Model	Best CV F_1m	Best Hyperparameters	Retrain Acc.	Retrain F_1m
PatchTST	.949	$d=128$, heads=16, layers=4, lr= $6.8e-5$, drop=.24	.965	.962
DeepONet	.941	rank=64, lr= $1.3e-3$, drop=.08, width=64	.968	.964
LSTM	.936	hidden=512, layers=3, drop=.23, lr= $2.4e-4$	.966	.961
FKM-AD	30 trials	$d_{model}=256$, rank=32, lr= $1.6e-3$, drop=.39	.967	.960

2 Ganymede Gas Production Forecasting

Dataset: 7 wells on the Norwegian Continental Shelf, 31,359 samples, 63 features. Input windows: 90 timesteps. Horizons: $h \in \{7, 14, 30, 90\}$ days. Outer split: temporal 80/20. Inner CV: 3-fold expanding window. Units: MMSCF (million standard cubic feet).

Table 3: Ganymede forecasting — held-out test (multi-well, MMSCF). Best values per column in **bold**. FM = foundation model (zero-shot).

Model	Metric	$h=7$	$h=14$	$h=30$	$h=90$
TimesFM ^{FM}	R^2_{prod}	.826	.763	.656	.413
	MAE	.740	.868	.949	1.02
TiRex ^{FM}	R^2_{prod}	.822	.775	.699	.565
	MAE	.658	.749	.804	.831
Chronos ^{FM}	R^2_{prod}	.734	.668	.584	.361
	MAE	.739	.835	.886	.995
PatchTST	R^2_{prod}	.574	.463	.177	−.718
	MAE	1.77	1.98	2.13	2.51
LSTM	R^2_{prod}	.327	.177	.213	−.464
	MAE	1.77	1.95	1.83	2.25
TCN	R^2_{prod}	−4.51	.108	.060	−.044
	MAE	3.98	2.37	2.62	2.34
DeepONet	R^2_{prod}	−5.72	−6.49	−26.8	−14.9
	MAE	4.84	4.99	5.81	5.70

Table 4: Ganymede held-out test — full metrics per model per horizon.

Model	Type	MAE				R^2_{prod}			
		$h=7$	$h=14$	$h=30$	$h=90$	$h=7$	$h=14$	$h=30$	$h=90$
TimesFM	FM	0.740	0.868	0.949	1.021	0.826	0.763	0.656	0.413
TiRex	FM	0.658	0.749	0.804	0.831	0.822	0.775	0.699	0.565
Chronos	FM	0.739	0.835	0.886	0.995	0.734	0.668	0.584	0.361
PatchTST	NN	1.771	1.980	2.134	2.510	0.574	0.463	0.177	−0.718
LSTM	NN	1.768	1.948	1.833	2.254	0.327	0.177	0.213	−0.464
TCN	NN	3.982	2.369	2.623	2.336	−4.509	0.108	0.060	−0.044
DeepONet	NN	4.836	4.987	5.812	5.700	−5.716	−6.495	−26.79	−14.93

Table 5: Ganymede 3-fold expanding window CV (multi-well, mean $\pm$ std). Top: R^2_{prod} , bottom: MAE.

Model	Metric	$h=7$	$h=14$	$h=30$	$h=90$
TimesFM ^{FM}	R^2	.69 $\pm$.07	.60 $\pm$.10	.48 $\pm$.17	.35 $\pm$.28
TiRex ^{FM}	R^2_{prod}	.74 $\pm$.07	.67 $\pm$.13	.60 $\pm$.17	.49 $\pm$.22
Chronos ^{FM}	R^2_{prod}	.63 $\pm$.05	.54 $\pm$.12	.43 $\pm$.19	.31 $\pm$.24
PatchTST	R^2_{prod}	.56 $\pm$.40	.39 $\pm$.60	.14 $\pm$.91	-.23 $\pm$ 1.34
LSTM	R^2_{prod}	.69 $\pm$.17	.54 $\pm$.34	.41 $\pm$.47	.20 $\pm$.57
TCN	R^2_{prod}	-7.96 $\pm$ 12.29	-1.07 $\pm$ 2.52	-3.52 $\pm$ 5.93	-6.68 $\pm$ 10.31
DeepONet	R^2_{prod}	-.03 $\pm$ 1.07	-.17 $\pm$ 1.16	-10.9 $\pm$ 16.4	-27.9 $\pm$ 40.2
TimesFM ^{FM}	MAE	3.54 $\pm$ 1.97	3.96 $\pm$ 2.19	4.28 $\pm$ 2.38	4.68 $\pm$ 3.45
TiRex ^{FM}	MAE	3.11 $\pm$ 1.75	3.34 $\pm$ 1.84	3.57 $\pm$ 1.97	3.99 $\pm$ 2.83
Chronos ^{FM}	MAE	3.42 $\pm$ 1.93	3.65 $\pm$ 2.02	3.93 $\pm$ 2.18	4.35 $\pm$ 2.97
PatchTST	MAE	4.16 $\pm$ 1.41	4.24 $\pm$ 1.33	4.56 $\pm$ 1.48	4.70 $\pm$ 1.88
LSTM	MAE	3.78 $\pm$ 1.88	4.08 $\pm$ 1.69	4.59 $\pm$ 1.96	4.81 $\pm$ 2.68
TCN	MAE	5.28 $\pm$ 0.72	5.05 $\pm$ 0.85	5.15 $\pm$ 1.10	5.79 $\pm$ 1.61
DeepONet	MAE	5.26 $\pm$ 1.56	5.36 $\pm$ 2.06	6.74 $\pm$ 0.36	7.50 $\pm$ 1.86

Table 6: Ganymede held-out test — RMSE and MASE per model per horizon (multi-well).

Model	Metric	$h=7$	$h=14$	$h=30$	$h=90$
TimesFM ^{FM}	RMSE	1.389	1.591	1.737	1.888
TiRex ^{FM}	RMSE	1.398	1.525	1.602	1.635
Chronos ^{FM}	RMSE	1.623	1.764	1.820	1.932
PatchTST	RMSE	2.192	2.441	2.717	3.307
LSTM	RMSE	2.616	2.876	2.546	3.090
TCN	RMSE	7.082	3.201	3.479	3.032
DeepONet	RMSE	7.910	8.196	13.360	9.686
TimesFM ^{FM}	MASE	1.681	1.012	1.117	1.415
TiRex ^{FM}	MASE	1.496	0.873	0.946	1.152
Chronos ^{FM}	MASE	1.679	0.974	1.043	1.379
PatchTST	MASE	4.025	2.308	2.511	3.479
LSTM	MASE	4.018	2.271	2.158	3.123
TCN	MASE	9.050	2.761	3.088	3.236
DeepONet	MASE	10.99	5.814	6.842	7.899

2.1 Per-Well Results

Table 7: Ganymede R^2_{prod} per well — $h=7$ (held-out test). Best per well in **bold**.

Model	Z01Z	Z02Z	Z04	Z05Z	Z06	Z07	Z08
TimesFM ^{FM}	-0.788	-0.342	0.153	0.001	0.279	0.208	-0.517
TiRex ^{FM}	-0.749	-0.248	0.129	0.179	0.205	0.117	-0.830
Chronos ^{FM}	-0.832	-0.467	-0.122	-0.050	0.114	-0.154	-1.767
PatchTST	-2266	-0.213	-0.345	-151.5	-7.139	-1.339	-0.825
LSTM	-173.8	-0.563	-5.823	-0.659	-1.337	0.012	-3.831
TCN	-48.04	-2.491	-9.858	-7.514	-6.702	-0.347	-8.158
DeepONet	-110.6	-1.307	-34.59	-112.8	-60.39	-0.447	-100.2

Table 8: Ganymede R^2_{prod} per well — $h=14$ (held-out test). Best per well in **bold**. TCN per-well results not available for $h=14$.

Model	Z01Z	Z02Z	Z04	Z05Z	Z06	Z07	Z08
TimesFM ^{FM}	-1.101	-0.822	-0.022	-0.579	-0.702	-0.050	-0.423
TiRex ^{FM}	-1.030	-0.714	-0.075	-0.213	-0.771	-0.155	-0.722
Chronos ^{FM}	-1.162	-0.877	-0.360	-0.469	-0.885	-0.498	-1.582
PatchTST	-3963	-0.579	-0.478	-187.1	-57.43	-1.964	-1.288
LSTM	-6.691	-0.394	-1.007	-1.107	-5.142	0.300	-0.996
DeepONet	-500.6	-4.950	-11.62	-87.01	-346.9	-1.203	-88.77

3 SPE BERG Gas Production Forecasting

Dataset: 53 anonymized horizontal multi-fractured wells (Eagle Ford unconventional gas). 67 features. Horizons: $h \in \{7, 14, 30, 90\}$ days. Inner CV: 3-fold expanding window. Units: Gas_Volume_MMscf.

Table 9: Ganymede R_{prod}^2 per well — $h=30$ (held-out test). Best per well in **bold**. TCN per-well results not available for $h=30$.

Model	Z01Z	Z02Z	Z04	Z05Z	Z06	Z07	Z08
TimesFM ^{FM}	-1.149	-1.288	-0.356	-1.323	-0.595	-0.260	-1.212
TiRex ^{FM}	-1.104	-1.188	-0.477	-0.693	-0.686	-0.408	-1.357
Chronos ^{FM}	-1.213	-1.398	-0.898	-0.834	-0.855	-0.764	-1.490
PatchTST	-4761	-0.826	-1.013	-244.1	-111.1	-3.556	-1.795
LSTM	-9.739	-1.154	-0.549	-4.927	-3.401	-0.250	-17.17
DeepONet	-487.3	-8.848	-19.63	-55.93	-101.2	-1.308	-710.3

Table 10: Ganymede R_{prod}^2 per well — $h=90$ (held-out test). Best per well in **bold**. TCN per-well results not available for $h=90$.

Model	Z01Z	Z02Z	Z04	Z05Z	Z06	Z07	Z08
TimesFM ^{FM}	-1.059	-1.629	-0.693	-2.690	0.088	-0.451	-1.385
TiRex ^{FM}	-1.064	-1.640	-0.716	-2.020	-0.032	-0.463	-1.384
Chronos ^{FM}	-1.115	-1.548	-1.450	-2.211	-8.270	-0.987	-0.885
PatchTST	-4372	-0.983	-1.137	-267.5	-991.2	-5.227	-2.315
LSTM	-58.83	-1.384	-1.801	-0.919	-66.27	-0.372	-1.422
DeepONet	-202.1	-6.284	-17.52	-26.02	-694.3	-0.603	-1.861

4 CDF Anomaly Detection

Dataset: Cumulative Distribution Function reconstruction task. Outer split: temporal holdout ($n_{\text{test}} = 864$). Inner CV: 3-fold. Metric: reconstruction error (lower is better).

Table 11: Ganymede MAE per well — $h=7$ (held-out test, MMSCF). Best per well in **bold**.

Model	Z01Z	Z02Z	Z04	Z05Z	Z06	Z07	Z08
TimesFM ^{FM}	0.025	1.846	0.531	0.264	0.353	1.485	0.707
TiRex ^{FM}	0.024	1.647	0.483	0.236	0.295	1.381	0.738
Chronos ^{FM}	0.024	1.701	0.516	0.253	0.299	1.457	0.838
PatchTST	4.435	2.201	0.878	4.336	4.311	2.302	1.786
LSTM	0.982	2.309	1.521	0.390	1.695	1.974	2.835
TCN	0.524	3.394	1.970	0.824	3.921	2.175	3.554
DeepONet	0.787	3.305	2.893	2.848	7.069	2.176	9.681

Table 12: SPE BERG forecasting — held-out test (multi-well, MMscf). Best values per column in **bold**. FM = foundation model (zero-shot).

Model	Metric	$h=7$	$h=14$	$h=30$	$h=90$
TimesFM ^{FM}	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...
TiRex ^{FM}	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...
Chronos ^{FM}	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...
PatchTST	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...
LSTM	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...
TCN	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...
DeepONet	R^2_{prod}	...	...	...	...
	MAE	...	...	...	...

5 Cross-Dataset Summary

Table 13: SPE BERG 3-fold expanding window CV (multi-well, mean $\pm$ std). Top: R^2_{prod} , bottom: MAE.

Model	Metric	$h=7$	$h=14$	$h=30$	$h=90$
TimesFM ^{FM}	R^2	...	...	...	...
TiRex ^{FM}	R^2_{prod}	...	...	...	...
Chronos ^{FM}	R^2_{prod}	...	...	...	...
PatchTST	R^2_{prod}	...	...	...	...
LSTM	R^2_{prod}	...	...	...	...
TCN	R^2_{prod}	...	...	...	...
DeepONet	R^2_{prod}	...	...	...	...
TimesFM ^{FM}	MAE	...	...	...	...
TiRex ^{FM}	MAE	...	...	...	...
Chronos ^{FM}	MAE	...	...	...	...
PatchTST	MAE	...	...	...	...
LSTM	MAE	...	...	...	...
TCN	MAE	...	...	...	...
DeepONet	MAE	...	...	...	...

Table 14: SPE BERG Friedman test summary per horizon. χ^2 : test statistic; p : p-value; ranks from lowest (best) to highest (worst) for R^2_{prod} and MAE.

Horizon	χ^2	p	Rank order (R^2_{prod} $\uparrow$)	Rank order (MAE $\downarrow$)
$h=7$	...	...	...	...
$h=14$	...	...	...	...
$h=30$	...	...	...	...
$h=90$	...	...	...	...

Table 15: CDF anomaly detection — held-out test reconstruction error. Lower is better. Best values in **bold**. FM = foundation model (zero-shot).

Model	Held-out Test					CV
	Mean	Std	P50	P95	P99	Mean $\pm$ Std
LSTM	0.005	0.003	0.004	0.009	0.014	0.099 ± 0.104
PatchTST	0.061	0.061	0.047	0.146	0.235	0.262 ± 0.137
DeepONet	0.219	0.123	0.186	0.449	0.503	0.975 ± 0.605
Chronos ^{FM}	24.81	19.19	19.66	60.92	82.89	24.83 ± 3.86
TimesFM ^{FM}	24.88	21.00	19.07	64.89	90.24	23.65 ± 3.65
TiRex ^{FM}	24.88	20.48	19.21	63.19	85.22	21.93 ± 3.22
<i>Mean baseline</i>	<i>41.46</i>	<i>17.51</i>	<i>39.68</i>	<i>66.26</i>	<i>93.90</i>	—

Table 16: Model performance summary across all four tasks. Arrows indicate metric direction ($\uparrow$ = higher is better, $\downarrow$ = lower is better). Best in **bold**. N/A = model not evaluated on task.

Model	3W (Classification)		Ganymede ($h=7$)		SPE BERG ($h=7$)		CDF
	Acc. $\uparrow$	F_1 m $\uparrow$	$R^2_{\text{prod}} \uparrow$	MAE $\downarrow$	$R^2_{\text{prod}} \uparrow$	MAE $\downarrow$	Error $\downarrow$
DeepONet	.968	.964	-5.72	4.84	...	...	0.219
LSTM	.967	.962	.327	1.77	...	...	0.005
Random Forest ^{Tree}	.966	.964	N/A	N/A	N/A	N/A	N/A
PatchTST	.965	.962	.574	1.77	...	...	0.061
FKM-AD (feat.)	.967	.961	N/A	N/A	N/A	N/A	N/A
FKM-AD (raw)	.949	.938	N/A	N/A	N/A	N/A	N/A
ConvTimeNet (feat.) [†]	.966	.962	N/A	N/A	N/A	N/A	N/A
ConvTimeNet (raw)	.158	.027	N/A	N/A	N/A	N/A	N/A
MambaSL (feat.) [†]	.966	.961	N/A	N/A	N/A	N/A	N/A
MambaSL (raw) [†]	.919	.887	N/A	N/A	N/A	N/A	N/A
TCN	N/A	N/A	-4.51	3.98	...	...	N/A
TiRex ^{FM}	.912	.895	.822	.658	...	...	24.88
TimesFM ^{FM}	N/A	N/A	.826	.740	...	...	24.88
Chronos ^{FM}	N/A	N/A	.734	.739	...	...	24.81

Table 17: Key findings per dataset.

Dataset	Winner	Key Finding
3W	Trained models & Random Forest	DeepONet 96.8%, LSTM 96.7%, FKM-AD 96.7%, RF 96.6%, MambaSL 96.6%, ConvTimeNet 96.6% RF, FKM-AD, MambaSL & ConvTimeNet prove feature extraction captures structure; Co
Ganymede	Foundation models	TiRex/TimesFM $R^2_{\text{prod}} > 0.8$ at $h=7$ TCN $R^2_{\text{prod}} \leq 0.11 \rightarrow$ local receptive fields insufficient
SPE BERG	...	... (Eagle Ford unconventional gas, 53 wells — results pending)
CDF	Trained models (task-specific)	LSTM error 0.005 vs FMs ~ 25 FMs lack CDF reconstruction inductive bias